

All Wired Up: An Exploration of the Electrical Properties of Microtubules and Tubulin

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Cite This: *ACS Nano* 2020, 14, 16301–16320



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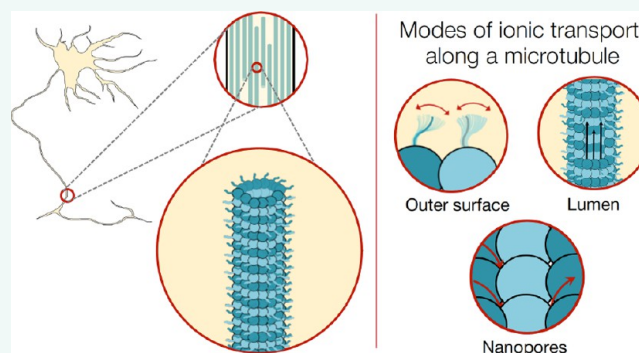
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ABSTRACT: Microtubules are hollow, cylindrical polymers of the protein α , β tubulin, that interact mechanochemically with a variety of macromolecules. Due to their mechanically robust nature, microtubules have gained attention as tracks for precisely directed transport of nanomaterials within lab-on-a-chip devices. Primarily due to the unusually negative tail-like C-termini of tubulin, recent work demonstrates that these biopolymers are also involved in a broad spectrum of intracellular electrical signaling. Microtubules and their electrostatic properties are discussed in this Review, followed by an evaluation of how these biopolymers respond mechanically to electrical stimuli, through microtubule migration, electro-rotation and C-termini conformation changes. Literature focusing on how microtubules act as nanowires capable of intracellular ionic transport, charge storage, and ionic signal amplification is reviewed, illustrating how these biopolymers attenuate ionic movement in response to electrical stimuli. The Review ends with a discussion on the important questions, challenges, and future opportunities for intracellular microtubule-based electrical signaling.

KEYWORDS: microtubules, tubulin, bionanowires, biosensors, intracellular charge transport, bioelectricity, intracellular signaling, cytoskeletal devices



INTRODUCTION

Microtubules (MTs) are slender pipelike polymers of the protein α , β -tubulin, that form a network of tracks for intracellular macromolecular transport. They establish morphological transitions during neuronal development and exert synchronized mechanical pulling forces for chromosomal segregation during mitosis. Interacting closely with actin filaments and intermediate filaments, MTs form a network of cooperating polymers known as the cytoskeleton, which provides mechanical support for the maintenance of cell shape and rigidity.

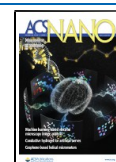
An MT has a cylindrical structure consisting of 13 vertically stacked columns (protofilaments) of tubulin dimers assembled in a head-to-tail manner, resulting in a 15 nm wide hollow core known as the lumen, and a 25 nm wide outer diameter (Figure 1a). Freely diffusing GTP (guanosine triphosphate)-bound tubulin dimers (GTP-tubulin) in solution are assembly competent and bind to a protofilament end, causing MT elongation. GTP-tubulin dimers that bind to the MT end are incorporated into the so-called MT lattice and eventually hydrolyze GTP bound to the β -monomer into GDP (guanosine

diphosphate) while GTP bound to the α -monomer does not hydrolyze to GDP. The hydrolysis of GTP-tubulin to GDP-tubulin is thought to induce mechanical strain causing a conformational change, which results in abrupt MT shortening (“catastrophe”). Catastrophes involve tubulin oligomers “peeling off” protofilament ends, sometimes leading to complete MT disassembly. Following a catastrophe, re-elongation events (“rescue”) restart MT growth until another catastrophe occurs, leading MTs to exist in a state of dynamic instability.^{27–29} Within a cell, dynamic instability is tightly regulated by families of microtubule-associated proteins (MAPs). The MAP tau, for example, is a wirelike protein that links several tubulin dimers within a protofilament or between neighboring MTs (e.g., in a

Received: August 18, 2020

Accepted: November 16, 2020

Published: November 20, 2020



ACS Publications

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16301

<https://dx.doi.org/10.1021/acsnano.0c06945>
ACS Nano 2020, 14, 16301–16320

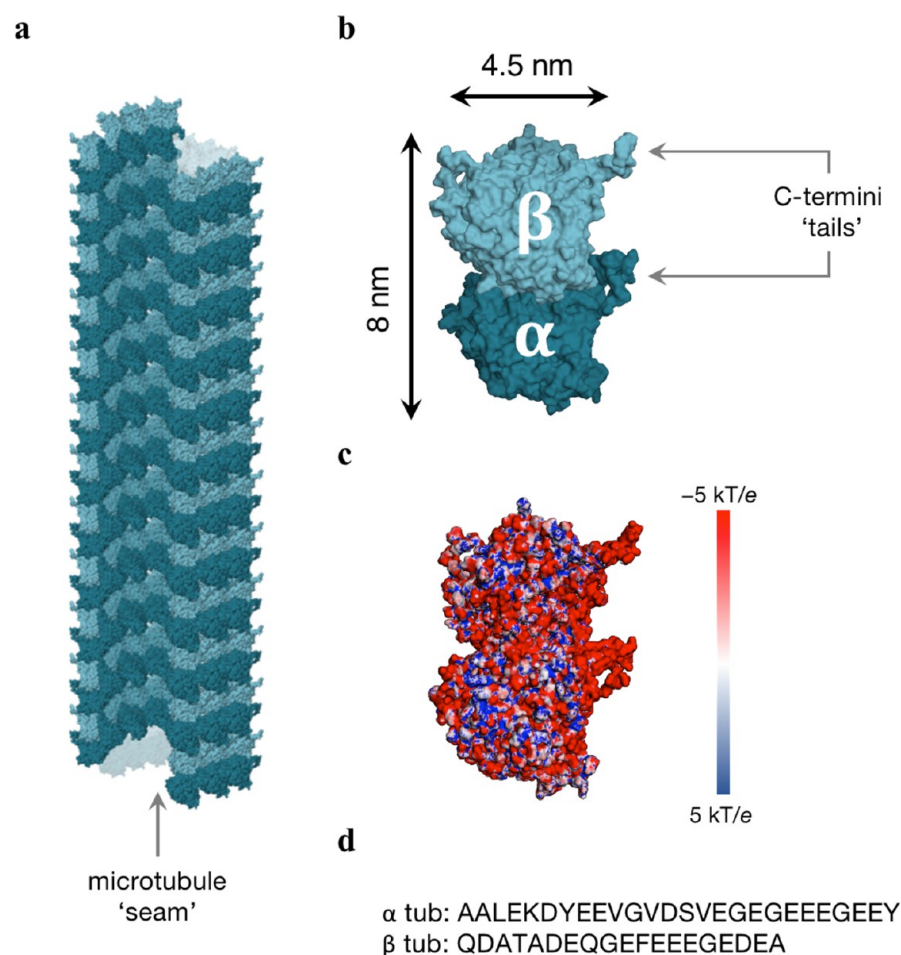


Figure 1. (a) Structure of MTs, displaying the cylindrical cross section and head to tail arrangement of tubulin dimers. (b) Structure of a tubulin dimer, showing C-termini tails, adapted from protein sequence 3RYF.⁶² (c) Electrostatic map of tubulin showing its dipole moment. Image created using APBS and PDB 2PQR software.^{18,63} (d) Amino acid sequence of the C-termini tails in the α (26 residues) and β (19 residues) subunits of tubulin as represented by homology modeling of 3RYF. In the 3RYF tubulin sequence, aspartic acid (D) and glutamic acid (E) together constitute 48.9% of the C-termini amino acid sequence.

neuronal cell) and stabilizes MTs against catastrophe.^{30,31} The MAP MCAK (mitotic centromere-associated kinesin), however, has the opposite function: MCAK molecules cause MT shortening by unbinding GTP-tubulin from the MT end.³² While MAPs control the ratio of polymerized and unpolymerized tubulin within the cell, it is worth noting that not all MAPs attenuate dynamic instability. Kinesin-1 (henceforth referred to as kinesin), for example, utilizes energy from ATP (adenosine triphosphate) hydrolysis to “walk” in a processive manner along the outer MT surface toward the plus (growing) end of an MT while transporting organelles such as mitochondria across the cell. While one end of kinesin (the motor “head” domain) interacts with tubulin and hydrolyses ATP, the other end (the “tail” domain) interacts with the cargo macromolecule.^{33,34} Several drugs and anesthetic molecules also bind to tubulin, mimicking the action of MAPs. The binding of the chemotherapeutic agent paclitaxel, for example, leads to dramatically increased MT nucleation and structural stability.^{16,35} The action of laulimalide, an investigational cancer chemotherapeutic agent,³⁶ also achieves a similar effect. On the other hand, the introduction of colchicine, which is used as an oral treatment for gout,³⁷ leads to inhibition of tubulin binding at the MT end, resulting in destabilization of the MT network.³⁸

Together, MAPs and drugs that interact with tubulin (collectively called tubulin binding agents, or TBAs) form a diverse set of chemical agents that modulate MT interactions with the rest of the cell.

Several intracellular roles of MTs, such as scaffolding, macromolecular cargo transport, and force generation for chromosomal segregation, rely on their mechanical properties: The persistence length of an MT is estimated to be between 1 and 10 μm ,³⁹ indicating that they are approximately 100 times stiffer than actin filaments and, especially, intermediate filaments,⁴⁰ while their longitudinal Young’s modulus is 1–5 GPa, similar to that of rigid plastics.^{41–43} The high stiffness of these biopolymers has prompted comparisons to carbon nanotubes (CNTs), which have comparable dimensions.⁴⁴ Similar to CNTs, MTs *in vitro* can be chemically conjugated to a variety of macromolecules, including synthetic proteins, quantum dots,⁴⁵ and magnetic nanoparticles.⁴⁶ The ease of labeling MTs with chemical agents, the intracellular abundance of tubulin, the high mechanical strength of MTs, the directionality of kinesin-based transport, and the ready formation of MTs at room temperature have prompted a thrust to develop highly controllable mechanically based MT nanodevices.^{39,47} Several transport devices that utilize MT-kinesin

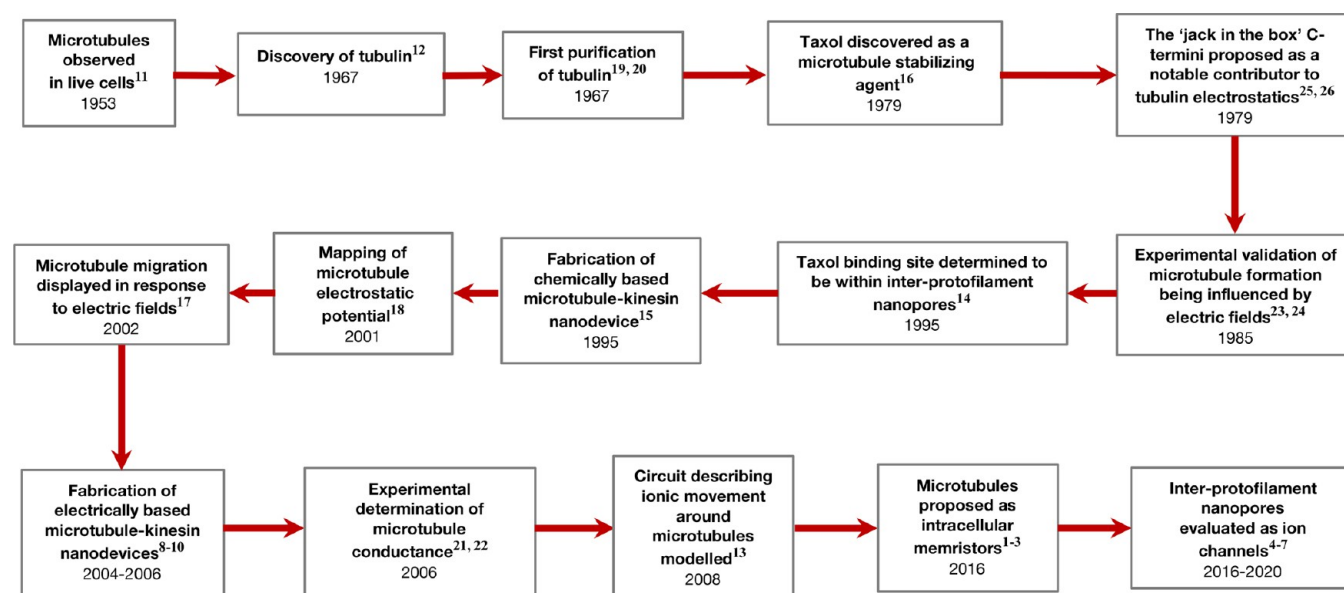


Figure 2. Timeline of relevant significant findings that contributed to present knowledge of the electrical properties of MTs.

systems, such as photolithography-based “rectifiers”^{10,48} and “concentrator” devices,⁴⁹ have already been fabricated, with the goal of overcoming obstacles posed by the “big fingers” of nanotechnological instrumentation,⁵⁰ (Figure 3c, d). However, due to the cumbersome nature of the lithographic process, optical and chemical techniques to control MT transport are also being explored.^{51–53} In the past few years, MTs have also found use as transporters on nontopographical surfaces,⁵¹ as three-dimensional microstructures such as bridges and pillars,^{54,55} and inside larger inorganic tubes (as a “tube within a tube”; Figure 3e).⁵⁶ In a “tube within a tube” the outer inorganic tubes have tunable dimensions, which present a promising potential as MT-based guides for directional transport. The mechanical strength of MTs has enabled their use in force-meters⁵⁷ and stiff substrates atop which CNT gliding can take place.⁵⁸ Larger MT-based superstructures (such as MT loops, asters and tori) have also been assembled, with potential use as nanopatterning agents (Figure 3f).^{59,60} Kinesin-driven MT transport has also been used to solve non-deterministic-polynomial-time (NP)-complete problems, which are encoded in an all-on-chip platform, providing a lower energy consumption and heat dissipation compared to other parallel computing approaches, including DNA computation, quantum computation, and microfluidic-based computation.⁶¹ The kinesin-MT system is thus viewed as a promising biodegradable (and potentially biocompatible) tool for the future of nanofluidics.

MTs have interesting electrical properties that have only recently begun to be utilized in molecular devices. Electrostatically, a tubulin dimer carries a bare (vacuum) charge of $-52 e$ under physiological pH (from the 3RYF structure), about 46% ($-24 e$) of which is localized at the tail-like C-termini attached to each monomer (Figure 1b, c). This charge is conferred by aspartic and glutamic acid residues, which together constitute approximately 49% of the C-termini amino acid sequences (Figure 1d). Due to this skewed charge distribution, each monomer acquires a permanent dipole moment of approximately 4000 D (Figure 1c). Within an MT, the radially symmetric arrangement of tubulin amplifies the protein dipole moment, especially its longitudinal component (Figure 1a),

causing MTs to acquire electrostatic polarity, predominantly along the MT axis, and a length-dependent net dipole moment.

Recent efforts to fabricate nanodevices that utilize the interesting and highly unusual electrical properties of tubulin show great promise for bionanotechnology.^{64,65} By comparison, a better understanding of the electrical properties of biomaterials such as melanin and DNA have enabled their utilization in devices such as OLEDs and solar cells.^{66,67} However, due to the lack of understanding of MT bioelectricity, extension from such devices has not been possible as of yet. However, MTs and tubulin have been modeled to act as targets for the action of electric-field based cancer treatment and wound healing modalities.^{68–72} Indeed, if the potential of MTs as biological analogues of CNTs is to be explored, a thorough examination of their responses to electrical stimulation will be required.

Historically, when they were imaged in live cells in 1953, MTs were considered important for chromosomal movement in cell division.¹¹ With findings from the 1960s and 1970s based on the successful *in vitro* purification of tubulin,^{19,20} showing that taxol had an MT stabilizing effect¹⁶ and that the C-termini tails were highly acidic,^{25,26} the electrical properties of MTs gradually came under focus. While there were indications that MTs and tubulin interacted in a nontrivial manner with external electric fields,^{23,24} the intracellular significance of their electrical properties and their device potential remained poorly understood. However, by the turn of the millennium, as nanofabrication and device miniaturization attained industrial relevance, MTs began being noticed as potential macromolecular transport devices.¹⁵ An all-atom simulation allowed the mapping of the electrostatic potential and showed the difference in electrostatic potential between the plus and minus ends of a MT.¹⁸ Subsequently, MTs have been experimentally and theoretically shown to play important roles such as charge storage and transport within physiologically relevant environments.^{13,73,74} While their high mechanical stability has led to their utilization in a variety of nanodevices, their interesting electrical properties have found use within sorters and direction-specific transport.^{9,10,64,65,75} More recently, MTs have been discussed as molecular targets for treatment modalities based on electric fields.^{68,71,72,76,77} In the future, undoubtedly more work

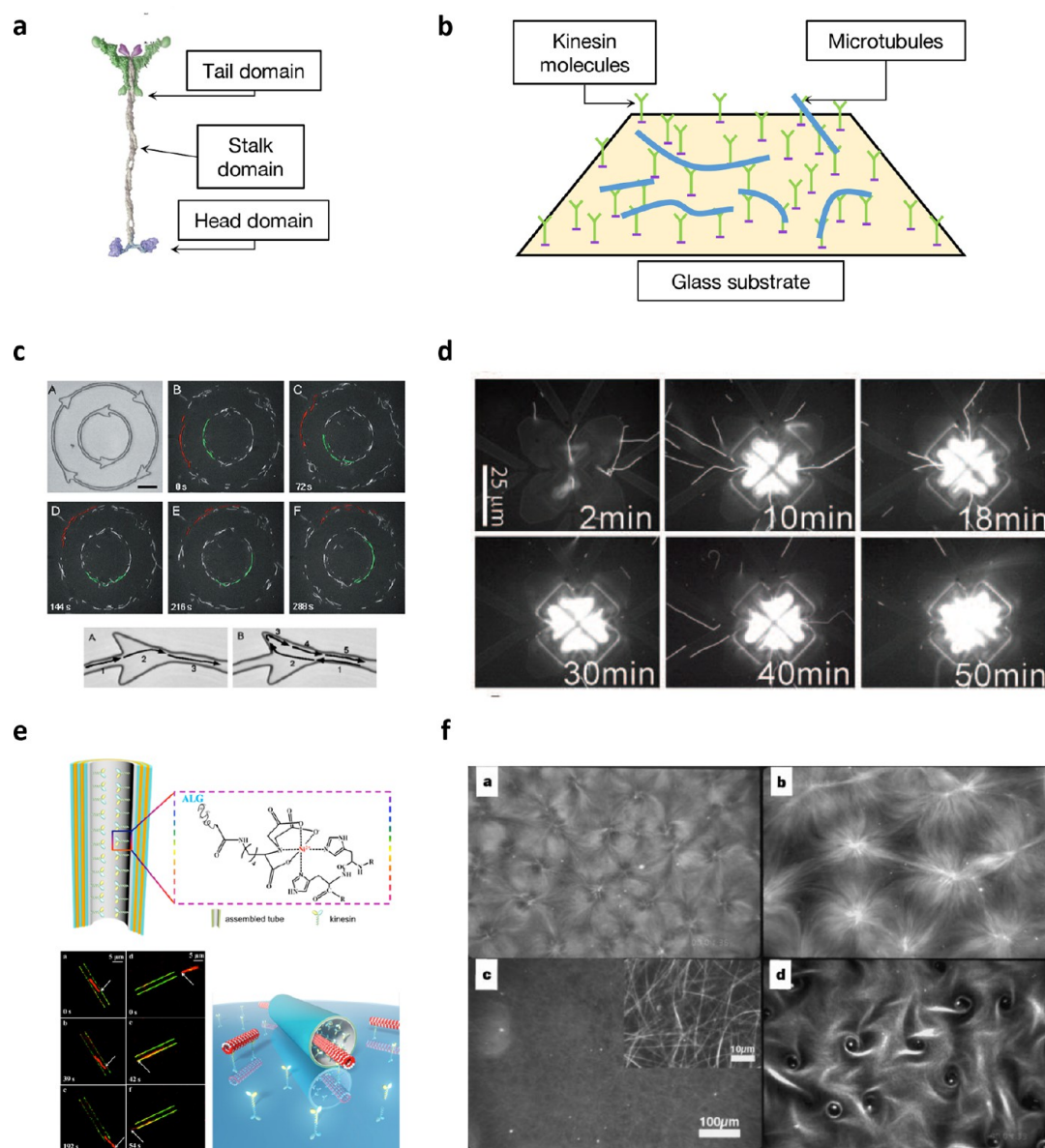


Figure 3. MTs as Nature's mesoscale devices. (a) Structure of kinesin, displaying the three functional domains. The head domain binds to tubulin, whereas the tail domain interacts with the cargo macromolecule to be transported. The head and tail domains are connected by the stalk domain. Reproduced with permission from ref 85. Copyright 2003 Elsevier B. V. (b) MTs (blue rod-shaped structures) are transported by the collective action of kinesins (green "Y" shaped structures). (c) MT-based rectifier device showing that the direction of MT gliding can be dictated using photolithographically etched, arrow-shaped patterns in the photoresist SAL600. Reproduced with permission from ref 48. Copyright 2001 Biophysical Society. (d) MT concentrator device can be used to focus MT gliding into a predetermined area through the use of several rectifier devices. Reproduced with permission from ref 49. Copyright 2008 American Chemical Society. (e) Transporting MTs as a "tube within a tube" through kinesin-coating on the inner wall of inorganic nanotubes. Reproduced with permission from ref 56. Copyright 2014 American Chemical Society. (f) Kinesin-MT interactions can lead to a variety of structures. Reproduced with permission from ref 59. Copyright 1997 Nature publishing group.

will be devoted to their utilization within nanodevices outside the cell, as well as their roles as targets for electrically based disease treatment. A timeline of milestones in research on the electrical properties of MTs is shown in Figure 2.

This Review showcases the state of the field in MT electrical properties, reviewing both applications within nanodevices as well as potential roles within the cell. While emphasis is placed on recent findings and illustrating the directions of future work, historically important work has also been pointed out to provide chronological perspective.

MICROTUBULES AS INTRACELLULAR SENSORS

The utilization of MT-based transport nanodevices has largely been realized through kinesin-conjugated migration. The macromolecule to be transported is first conjugated to kinesin, after which perfusion of the macromolecule-kinesin construct onto an MT-coated substrate is performed, causing kinesin's processive movement or "gliding" on MT tracks.^{60,78–80} A reversed configuration can also be used, in which the substrate is coated with kinesin molecules attached by their tail domains, while their MT-interacting head domains are exposed to solution (Figure 3a, b). Upon the perfusion of MTs into the

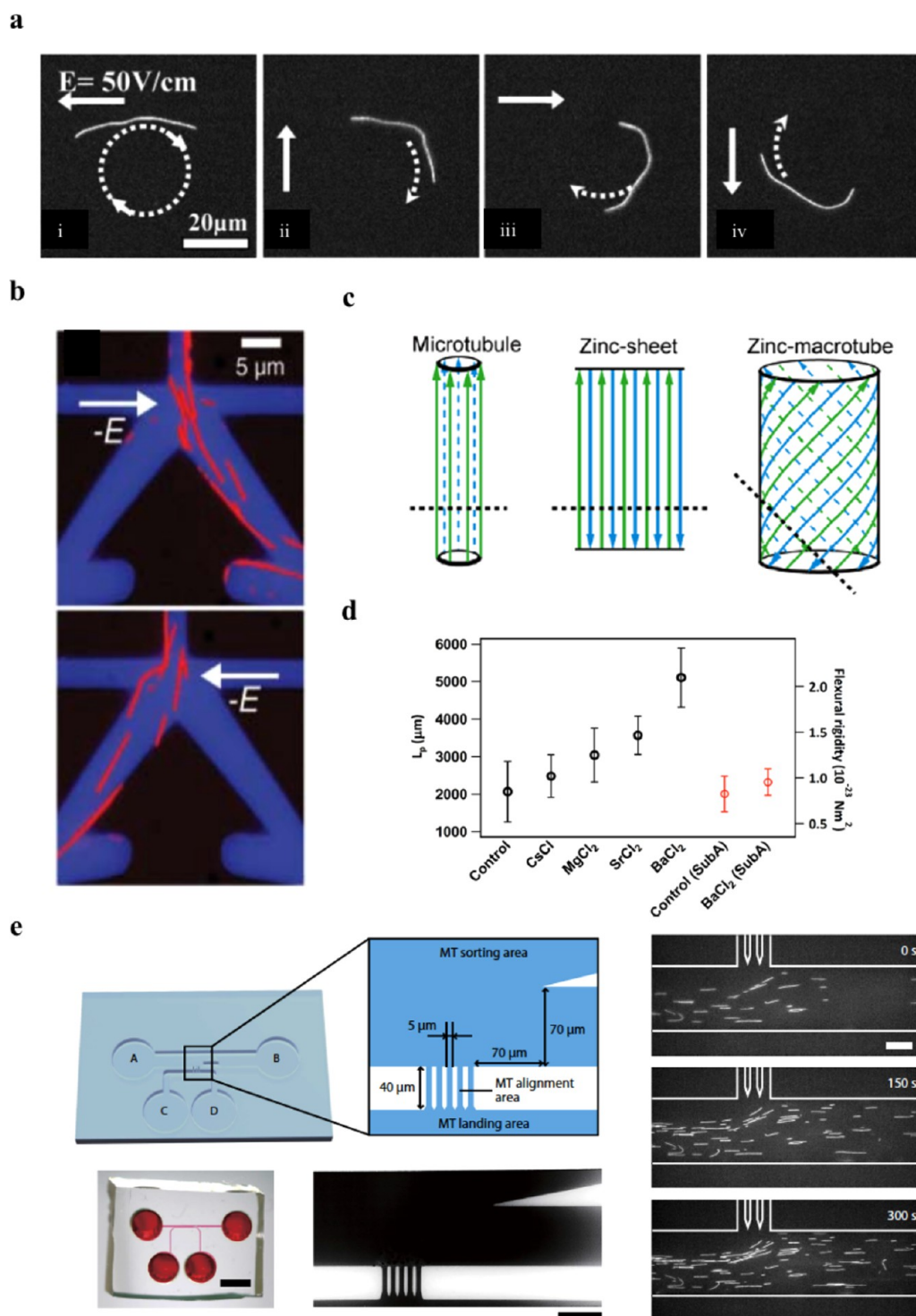


Figure 4. (a) Demonstration of circular translocation of a MT on a kinesin coated substrate using an electric field of 50 V/cm in directions specified by the solid arrow. Adapted with permission from ref 109. Copyright 2007 American Chemical Society. (b) MT steering on kinesin-coated microchannels with an electric field of 500 V/cm. Reproduced with permission from ref 9. Copyright 2006 American Association for the Advancement of Science. (c) Schematics of tubulin polymorphs, indicating MTs (left), sheets (center), and macrotubes (right). Reproduced with permission from ref 110. Copyright 2012 PLOS One. (d) Graph displaying the variation of persistence length on the addition of various divalent salts. Reproduced with permission from ref 95. Copyright 2014 American Chemical Society. (e) Sorting device for kinesin steering MTs of varying stiffness. Varying curvature trajectories are achieved with a 30 V/cm field strength. Reproduced with permission from ref 64. Copyright 2017 American Association for the Advancement of Science.

chamber, the exposed head domains bind to tubulin, causing ATP to hydrolyze and kinesin to “step” forward. This leads to MT movement on the kinesin-coated substrate, eventually resulting in the desired macromolecular migration.

Irrespective of its configuration, each MT moves in a random direction governed by the Brownian motion of its tip on a kinesin-coated substrate. Thus, transport of cargo in one specific direction is hard to achieve and has been an area of focused

research over the last two decades. The application of an external electric field resolves this hurdle, causing the highly negative MT tip to orient along electric field lines, causing mechanical deflection and migration of the entire MT.^{8,10} This behavior was utilized to create rectifiers capable of transporting gold nanowires across tens of micrometers⁸ and sorting devices with accuracies in excess of 90% for kinesin-driven MTs steering under one-dimensional electric fields (Figure 4a, b).^{9,75} The use of MT electrostatics to tune their mechanical response in accordance with the requirements of the nanosystem adds immense versatility. For example, the electrostatic charge on an MT tip was successfully controlled by labeling it with double-stranded DNA, the presence of which resulted in significantly different radii of curvature, depending on the size of DNA used for labeling.⁶⁵ A recent study created an MT-based sorting device on a kinesin-coated glass substrate to study MTs polymerized in the presence of a slowly hydrolyzable GTP-analogue (GMPCPP⁸¹), which are known to be stiffer than those polymerized in the presence of GTP (Figure 4e). The authors subsequently managed to sort a mixture of GMPCPP and GTP polymerized MTs using nonuniform electric fields, based on the extent to which they bent.^{64,65} To explore the potential of electrically based MT transport devices, a deeper understanding of the response of kinesin processivity in the presence of a wide variety external electric field parameters is required.

Device fabrication based on the mechanical response of MTs to electric fields has also enabled insights into their electrical properties (such as electrophoretic mobility and zeta potential) which have been shown to depend strongly on buffer conditions (pH and ionic strength).¹⁷ Due to extensive counterionic condensation on the MT surface, the effective charge per tubulin (including surfaces that are incorporated into the MT wall) when it is within the MT lattice, is much lower than the bare tubulin charge of $-52 e$ at pH 6.9. Measurements of the electrophoretic mobility of MTs (values shown in Table 1) have allowed for calculation of the effective charge of a dimer in an MT to be $-23 \pm 2 e$ under physiological conditions.⁷⁵ Earlier publications reported measured tubulin mobilities (values also shown in Table 1) but concluded that the effective charge of tubulin was only $-12 \pm 2 e$ ⁹ or even as low as $-0.2 e$ ^{17,82} under physiological conditions. However, the calculation of $-12 \pm 2 e$ uses a simplistic model for the MT geometry and the calculation of $-0.2 e$ neglects the retardation effect (the additional drag force caused by positive counterions around, but not condensed on, the MT) and considers MT motion to be hydrodynamic.^{17,82} Therefore, counterionic condensation should be considered to screen roughly half of the bare charge of an MT. Interestingly, the effective charge is higher when a tubulin dimer is within the MT lattice as opposed to when it is in solution, indicating that tubulin polymerization alters its electrostatic charge distribution and that free tubulin in solution is electrically distinct from that embedded within the MT lattice (as polymerized tubulin). Also, the isoelectric point of tubulin within an MT was measured to be pH 4.2;¹⁷ however, previous experiments showed that the isoelectric point of free tubulin is slightly above pH 5.^{83,84} Such findings indicate the importance of the tubulin polymerization state, *i.e.*, free tubulin as opposed to that within an MT lattice (polymerized tubulin) on its biophysical characteristics. Future work will quantify how the polymerization state of tubulin changes the concentration of free ions and the electrical properties of a solution containing tubulin.

Table 1. Electrical Parameters of Tubulin and MTs Determined from the Mechanical Response to Electrical Stimuli^a

quantity	value
tubulin bare charge at pH 6.9	$-52 e$
tubulin effective charge in MT	$-23 \pm 0.2 e$ ⁷⁵
relative permittivity of tubulin	3.5251 ± 0.861 ⁸⁶
polarizability of tubulin	$7 \times 10^{-34} \text{ C} \cdot \text{m}^2/\text{V}$ ⁸⁶
effective tubulin dipole moment in MT (MT length $5 \mu\text{m}$, along the MT length)	2.8 D ⁸⁷
overall MT induced dipole moment (MT length $3.5 \mu\text{m}$; along the MT length)	$340\,000 \text{ D}$ at $200\,000 \text{ V/m}$, 2 MHz ⁸²
MT electrophoretic mobility (bulk value)	$(-2.6 \pm 0.4) \times 10^{-8} \text{ m}^2/(\text{V s})$ ⁸²
MT electrophoretic mobility (MT oriented parallel to the E-field)	$(-2.59 \pm 0.02) \times 10^{-8} \text{ m}^2/(\text{V s})$ ⁷⁵
	$(-2.93 \pm 0.02) \times 10^{-8} \text{ m}^2/(\text{V s})$ ⁹
MT electrophoretic mobility (MT oriented perpendicular to the E-field)	$(-2.15 \pm 0.01) \times 10^{-8} \text{ m}^2/(\text{V s})$ ⁷⁵
	$(-2.30 \pm 0.04) \times 10^{-8} \text{ m}^2/(\text{V s})$ ⁹
MT zeta potential	$-32.6 \pm 0.3 \text{ mV}$ ⁷⁵ $-43.5 \pm 6.5 \text{ mV}$ ⁹
MT effective linear charge density	$-280 e/\mu\text{m}$ ¹⁷ $-256 e/\mu\text{m}$ ⁸⁸
MT effective surface charge density	$-0.229 \pm 0.003 e/\text{nm}^2$ ⁷⁵ $-0.24 \pm 0.04 e/\text{nm}^2$ ²⁹

^aIonic concentrations were calculated assuming total dissociation of Mg^{2+} , Na^+ , K^+ , and Cl^- ions. All measurements were taken between pH 6.8 and 7.0.

A multitude of mechanical responses to electrical signals result from the length-dependent dipole moment of an MT. Early indications of this dipole moment came when, upon exposing a solution containing unpolymerized tubulin dimers to ac electric fields (10 Hz , 25 mV/cm), MT formation along electric field lines of force was observed.²⁴ The existence of a permanent dipole moment was confirmed when MTs that were initially oriented in random directions on a glass substrate reoriented when exposed to dc electric fields.⁸⁷ Successful electrorotation of MTs in the presence of ac electric fields (200 kHz to 2 MHz , $<2.1 \times 10^5 \text{ V m}^{-1}$) highlighted the significance of induced dipoles present within an MT. Induced dipoles were reported in response to external electric fields as resulting from a shift either in the electron densities in the tubulin dimer or in the counterion distribution condensed around the MT surface. Thus, an induced dipole as well as the MT's permanent dipole need to be considered when analyzing motion where alternating electric fields are present in the sample with tubulin or MTs.

Many of the interesting electrical properties of MTs are a result of the brushlike C-termini tails, which are about 4 nm in length (when fully stretched out) and protrude from the outer MT surface. At physiological pH values, each tail adopts a stretched random coil conformation, extending outward into the solution, due to electrostatic repulsion among their constituent negatively charged amino acids (Figure 6a, b⁸⁹) and away from the mainly negatively charged surface of an MT. As the solution's pH is lowered, amino acid protonation leads to a decrease in the absolute charge, causing the tail to bend inward with parts of it forming helical motifs.^{90,91} Even local temperature changes can supply enough thermal energy to cause C-termini conformational changes; these result in the C-

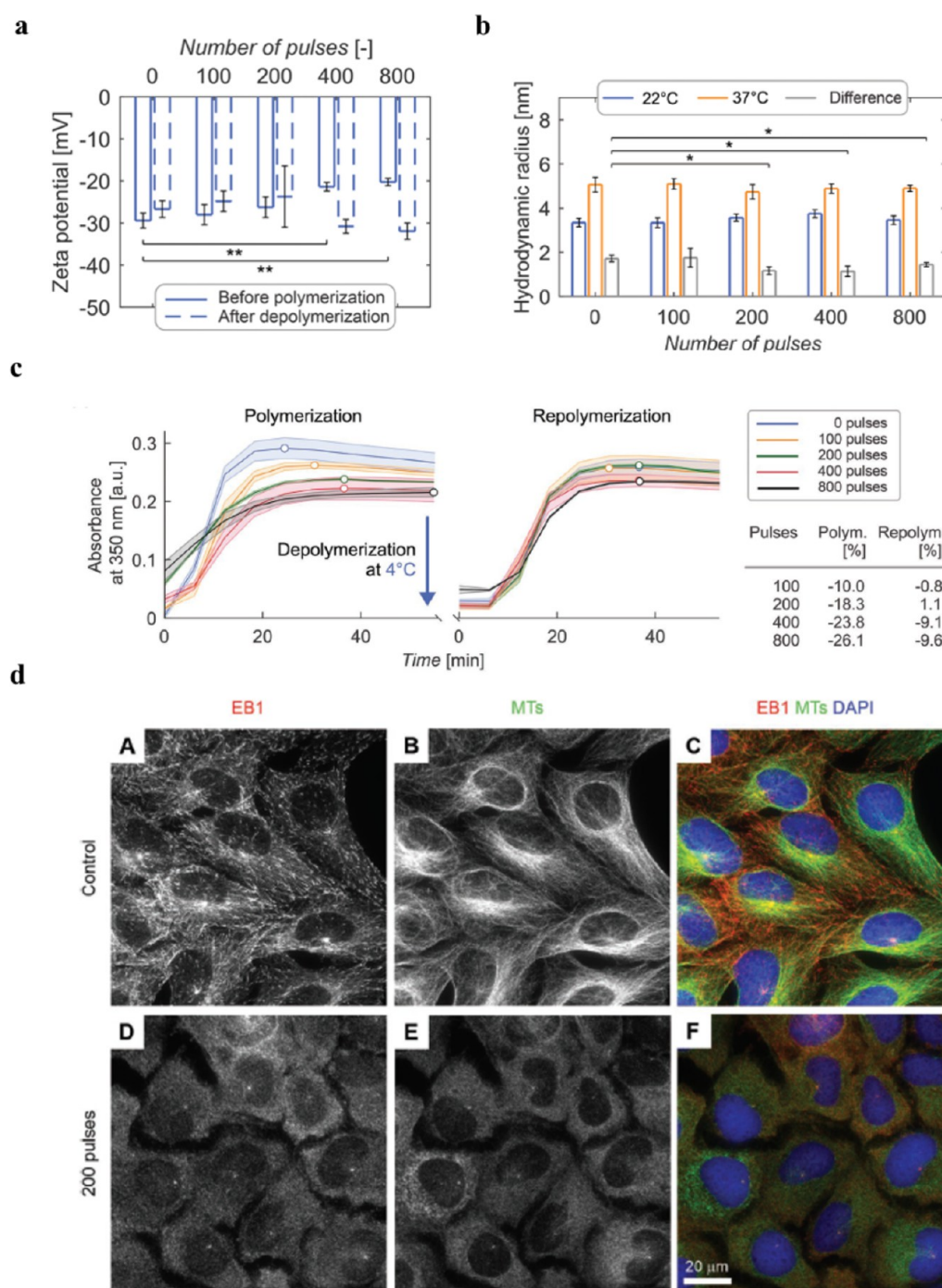


Figure 5. (a) Graphs displaying the changes in tubulin zeta potential values upon exposure to increasing number of nanosecond electropulses (nsEPs). Reproduced with permission from ref 71. Copyright 2019 Wiley-VCH. (b) Graphs displaying values of tubulin hydrodynamic radius upon incubation at different temperature values, and number of nsEPs. Reproduced with permission from ref 71. Copyright 2019 Wiley-VCH. (c) Turbidity experiments showing MT polymerization is inhibited in the presence of nsEPs (right). After MT depolymerization by cooling to 4 °C, exposure to nsEPs also resulted in inhibited repolymerization (left). Reproduced with permission from ref 71. Copyright 2019 Wiley-VCH. (d) Effect of nsEP exposure on U2OS cells imaged using immunofluorescence staining (top). Effect on the MAP, EB1 (end-binding protein 1; red), MTs (green), and nuclei (blue) can be clearly seen. Reproduced with permission from ref 72. Copyright 2020 Wiley-VCH.

termini mechanically oscillating like a “jack-in-the-box.”²⁶ The presence of C-termini increases electrostatic repulsion between (otherwise less negative) neighboring tubulin dimers, lowering MT elongation rates. At lower pH values, favoring of the kinked conformation and amino acid protonation results in a reduction in the electrostatic repulsion between neighboring tubulin causing an enhancement of MT elongation.⁹²

The addition of cations to the solution can regulate the material properties (including stiffness, morphology, and stability) of tubulin polymers through nonspecific interactions. For example, the presence of trivalent cations Al^{3+} and Fe^{3+} at high concentrations ($>250 \mu\text{M}$) causes MT destabilization and tubulin aggregation.^{93,94} The presence of Ba^{2+} , on the other hand, leads to an increase in MT stability and stiffness (e.g.,

increasing the persistence length from 2 μm in control conditions to 5 μm in the presence of 5 mM BaCl_2 ; Figure 4d).⁹⁵ The presence of Mg^{2+} and Mn^{2+} also increases MT stability while additionally leading to the formation of GDP-tubulin “rings” in solution.⁹⁶ Ba^{2+} , Mg^{2+} , and Mn^{2+} are modeled to interact nonspecifically with the tubulin dimer, lowering the electrostatic repulsion between the C-termini tail and tubulin body. The shielding effect is modeled to bias the C-termini toward a kinked conformation, stabilizing the overall MT structure.⁹⁷ It is worth noting that not all divalent cations have this effect; the presence of Zn^{2+} or Co^{2+} leads to the formation of polymeric, two-dimensional tubulin sheets instead of cylindrical MTs.^{98–100} Zn^{2+} interacts at the N-terminal of each tubulin monomer, causing protofilaments to align in an antiparallel orientation to one another (Figure 4c).¹⁰¹ Prolonged exposure to a solution of Zn^{2+} can lead to sheets “wrapping up” to form >300 nm diameter macrotubes, indicating the importance of electrostatics in the determination of the polymorphic structure of tubulin.^{99,102} Interestingly, Ca^{2+} has the opposite effect, causing MT disassembly through acceleration of GTP hydrolysis.^{103,104} Addition of monovalent cations Na^+ and K^+ stimulates MT assembly until a saturating concentration is reached, after which an inhibitory effect is observed.¹⁰⁵ The presence of Na^+ has also been observed to stabilize tubulin sheets for long durations of time.¹⁰² In contrast, high concentrations of the organometallic cation CH_3Hg^+ cause MT disassembly through potential interactions with free sulfhydryl groups on the tubulin surface.^{106,107} The presence of large (trivalent, tetravalent and pentavalent) cations derived from lysine oligomers induces the formation of hexagonal bundles of MTs and so-called “living necklaces”.¹⁰⁸ Collectively, these results indicate the potential utility of electrostatic parameters to enable the tuning of cargo volume, movement and direction within integrated nanodevices that utilize MT-kinesin systems for transport.

In addition to responding to the presence of different ions, C-termini tails are also modeled to undergo conformational changes when exposed to nanosecond electrical pulses (nsEPs; 20 kV/cm, >400 ns). Exposure to nsEPs has been experimentally shown to inhibit the ability of tubulin to form MTs (Figure 5a).⁷¹ Upon nsEP exposure, changes in zeta potential and hydrodynamic radii that are consistent with C-termini extension were clearly observed (Figure 5b and c). nsEP exposure to U2OS cells in a low conductivity buffer background led to MT loss, as seen by the loss of filamentous structures upon performing confocal microscopy (Figure 5d; center panel). This also resulted in fluorescently tagged end binding protein 1 (EB1), which can be imaged as distinct “comets” that localize to growing MT ends,^{111–113} which have been seen to instead diffusely spread across the cytoplasm (Figure 5d; far left panel).⁷² Other work has also shown that nsEP exposure with similar parameters led to intracellular MT disruption.^{114,115} Interestingly, a reversal of this MT destabilizing effect was seen in high conductivity media, where nsEP exposure led to a smaller number of MTs in cells overall, which underwent longer growth times. The nsEP-MT interaction is hence proposed as a potential target for nonpharmacological therapies (or in combination with pharmacological therapies) in the context of cancer and illustrates the utility of using electric fields to successfully induce biochemical changes within intracellular proteins. Importantly, already developed cancer treatment modalities involving electric fields such as tumor-treating fields (TTFields) are also hypothesized to interact mechanochemi-

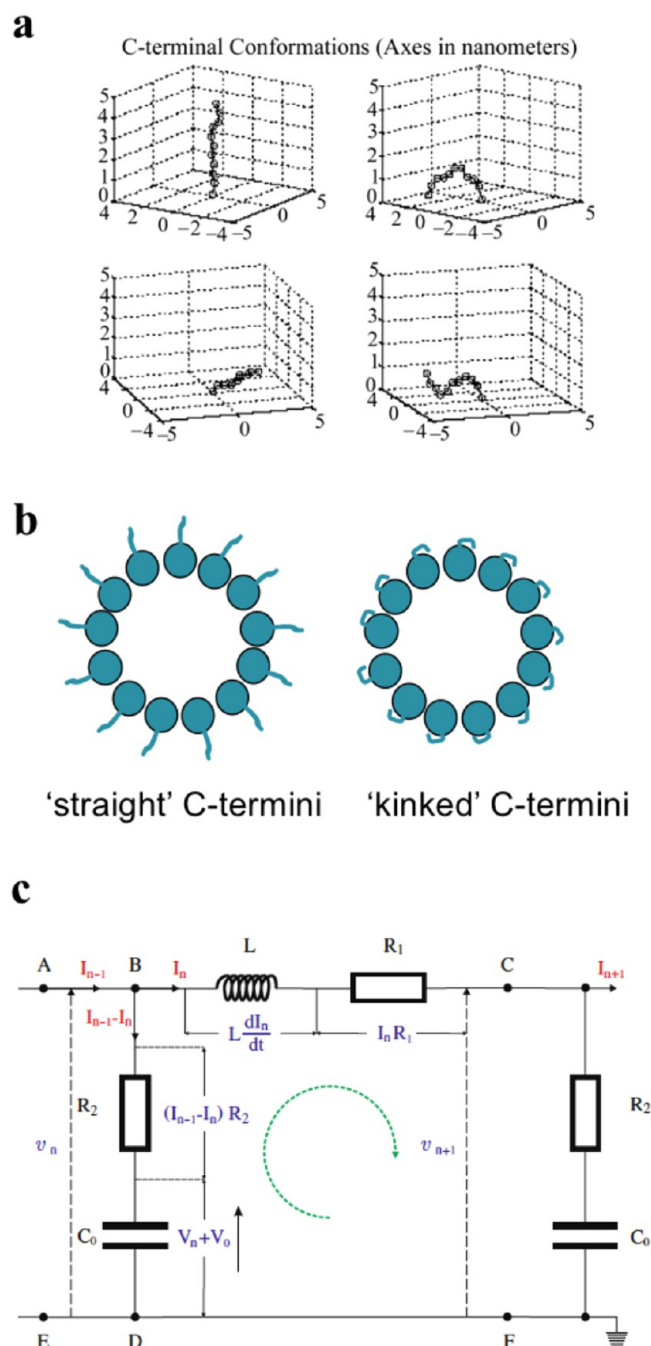


Figure 6. (a) Results from a simulation depicting various conformations adopted by a single C-terminus. Reproduced with permission from ref 74. Copyright 2005 Springer Nature. (b) Schematic of C-termini in the “straight” and “kinked” conformations, viewed from the MT cross section. (c) Equivalent circuit of ionic flow along the n th 8 nm section in a MT using a nonlinear transmission line model. Reproduced with permission from ref 119. Copyright 2009 Springer Nature.

cally with the MT network of the cancer cell.^{68–70} According to the standard model, TTFields, which are 10–300 kHz, 1–3 V/cm ac electric fields, inhibit cancer-cell proliferation by interacting with the large dipole moment of tubulin, which causes MT alignment along TTField-generated electric field lines leading to a disruption of the mitotic spindles. It is important to note that the precise molecular mechanism through which TTFields, or any other electric fields for that

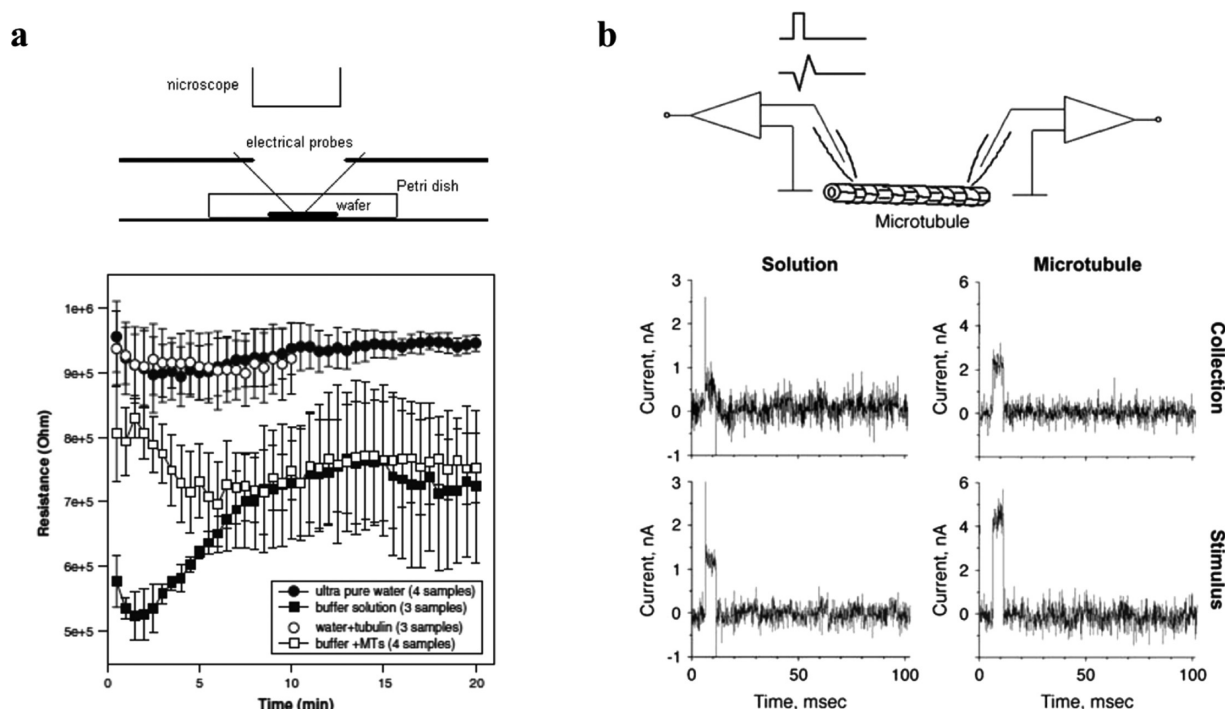


Figure 7. (a) Schematic representation of a probe-type geometry used to evaluate the dc response of MTs to voltage exposure (top). Variation of resistance of various samples as a function of time, measured at the probe tips (bottom). Reproduced with permission from ref 122. Copyright 2007 Springer Nature. (b) Experimental setup displaying patch-clamp based MT contact method (top). Variation of “stimulus” and “collection” currents as a function of time, indicating the potential of MTs as amplification devices within the cell (bottom). Reproduced with permission from ref 123. Copyright 2006 Biophysical Society.

matter, act on MT dynamic instability parameters has not yet been clearly established. If *in vitro* assays determining the relationship between electric field parameters (including frequency, intensity, and waveform type) could be performed, then the sensitivity and specificity of electric field based anticancer modalities would be significantly improved, providing an important piece of information needed to optimize the delivery of this therapeutic modality for cancer patients. Notably, upon translating the laboratory measurement results for MTs to those within living cells, an important difference involves the presence of cell membranes. It is well-known that the phospholipid bilayer structure of the membrane can be represented as a capacitor, which could be seen as a high-pass filter, which would allow sufficiently high frequency electric fields to permeate into the cytoplasm as is in evidence, when examining the action of TTFs.¹¹⁶ Furthermore, cell membranes incorporate numerous membrane proteins (including those that make up ion channels). This results in the electrical representation of the membrane as a “leaky capacitor” further allowing the electric fields to penetrate into the cell interior.¹¹⁷ Dividing cells, in particular, can be characterized by the presence of the cleavage furrow during cytokinesis, which additionally provides areas of electric field penetration.¹¹⁸ Therefore, the electric field effects of MTs in living cells become not only frequency dependent but also dependent on MT location within the cell.

MICROTUBULES AS INTRACELLULAR NANOWIRES

Due to their large negative charge, MTs cause extensive cationic condensation on their outer surface. The 4 nm long, brushlike C-termini tails, that protrude from the outer MT surface, increase the surface area available for cationic condensation. While

counterions closer to the MT surface are modeled to be “frozen”, those further away from the surface can move along the MT relatively freely, creating an axial conductive sheath along the MT.¹³ Some resistance to this ionic current is caused by thermal noise and interactions with the protein surface, with axial and perpendicular resistances for a single 8 nm section of an MT having been estimated to be 10^9 and $7 \times 10^6 \Omega$, respectively.¹¹⁹ A capacitance is predicted to be established by the presence of a theorized “depleted layer” with a thickness corresponding to the Bjerrum length (room temperature Bjerrum length in aqueous buffer = $6.7 \text{ \AA}$),¹¹⁹ which separates the condensed counterions from the bulk solution.^{119,120} The existence of this depleted layer still needs to be experimentally confirmed. However, if this model is correct, then the MT capacitance will be nonlinear due to C-termini dynamics and saturable size.¹²⁰ The capacitance of 8 nm section of an MT, excluding the C-termini, is predicted to be 0.68 fF, and the net capacitance due to all corresponding C-termini is estimated as 0.64 fF.¹²⁰ The helical arrangement of tubulin dimers in an MT is theorized to cause the axial currents to twist, forming a solenoidal path with a corresponding inductance of 0.8 fH per ring. These values of resistance, capacitance, and inductance, as well as the resistance to ionic flows through the nanopores (see [Ionic Currents across the MT Wall](#)), can be used in an equivalent circuit model to predict the behavior of ionic transmission along, around and inside an MT. Consequently, the complex geometry of an MT results in highly unusual ionic electric conduction behavior, which, when experimentally validated, can serve as a basis for the construction of complex nontrivial bioelectronic devices.

Unsurprisingly, MTs have been theorized to behave as transmission lines, with signals propagating across their lengths as nonlinear ionic waves.^{89,119,120} These waves can be described

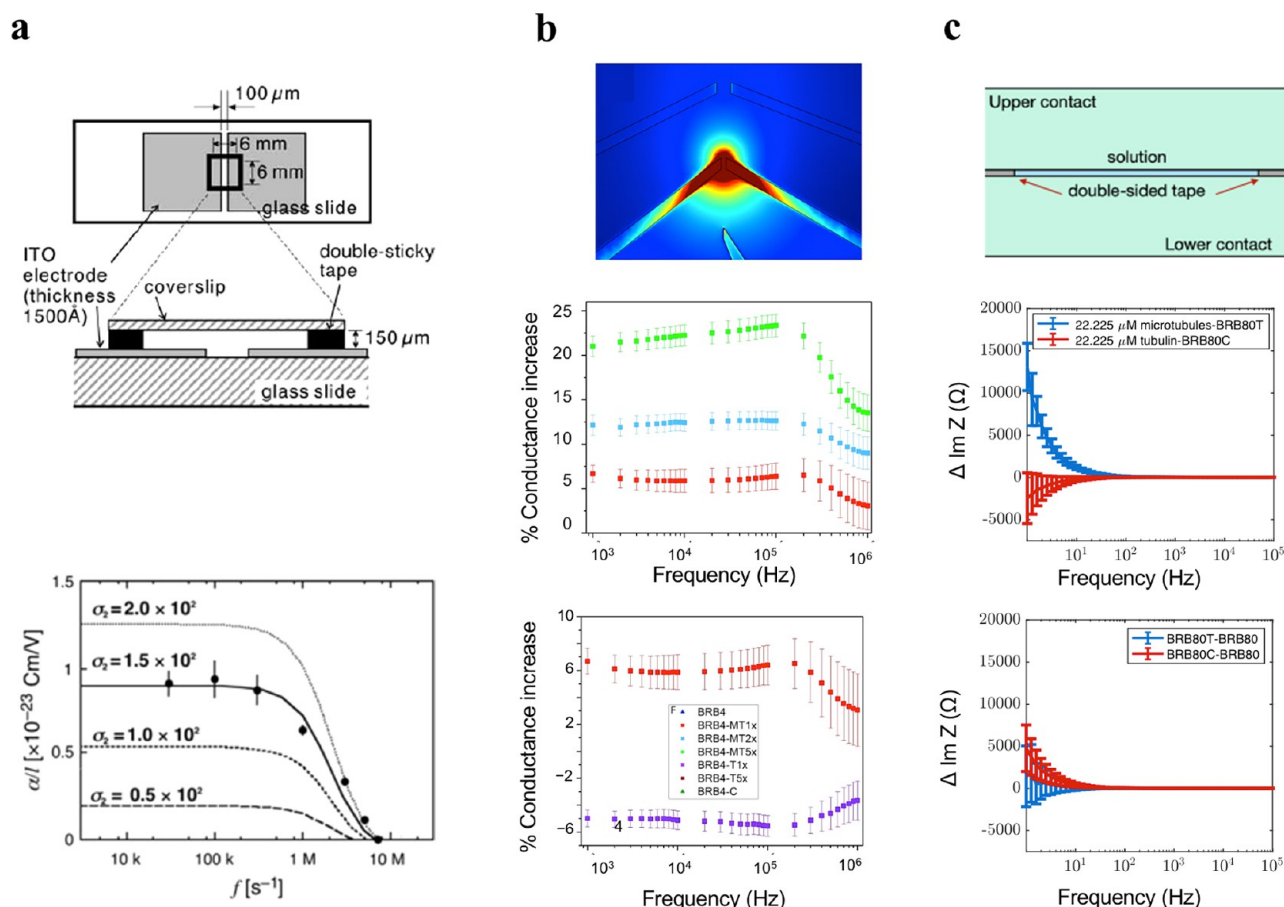


Figure 8. (a) Experimental setup used to determine MT conductivity through electrorotation experiments (top). Variation of polarization coefficient (α) per unit length, as a function of input electric field frequency. The best fit for the line for conductance is displayed to be 1.5×10^2 mS/m (bottom). Reproduced with permission from ref 22. Copyright 2006 Biophysical Society. (b) Spatial variation of electric field intensity of a probe-type contact geometry used to determine the effect of MTs and tubulin on solution conductance (top). Variation of MT and tubulin conductance as a function of frequency (below). Adapted with permission from ref 76. Copyright 2017 Springer Nature. (c) Schematic representation of a parallel-plate contact geometry used to determine the effect of MTs (at cell-like concentrations of tubulin and ions) on solution capacitance (top). Differences between imaginary components of tubulin and MTs, with their respective background solutions (below). Adapted with permission from ref 127. Copyright 2020 Multidisciplinary Digital Publishing Institute.

by their associated potential, which was shown to have an antikink wavefront along the MT length.¹¹⁹ For reasonable circuit parameter values, monovalent and divalent counterionic waves were shown to propagate as solitons, which retain their localized waveform despite experiencing ohmic losses due to the presence of nonlinear effects. These soliton pulses have propagation speeds estimated to be in the mm/s to cm/s range.^{120,121} The sheer magnitude of ionic wave velocity may help explain how subcellular electrical signals can propagate amidst large amounts of thermal noise in an aqueous environment. The influence of MTs on local charge carrier dynamics, through the complex interconnected network formed within cells, particularly neurons, highlights their potentially important role in bioelectric signal transmission.

Ionic Currents along the MT Outer Surface. Early hints that MTs altered ionic transport came in 1985, when Vassilev and co-workers showed that the presence of MTs led to electrical coupling between two lipid bilayer membranes (LBMs) that were separated by a 100 μm gap.²³ As the interesting electrical properties of the C-termini and counterionic condensation around MTs were revealed, several experimental efforts to evaluate counterionic conductivity around MTs were performed. However, the determination of

the ionic conductance around a single MT is a major experimental challenge, mainly because the equivalent circuit formed by thousands of cylindrical MTs of different lengths (each with its own resistive, capacitive, and inductive components) oriented in random directions in a background solution is difficult to reliably reconstruct. Reliable experimental determination of MT conductivity is further hampered by the high conductivity and experimental noise associated with high ionic strength buffers (such as BRB80) in which MTs are most stable. Despite such obstacles, several noteworthy attempts at determining the MT conductivity have been made and are discussed below.

To quantify the dc conductivity of an individual MT from an ensemble, the electrostatic interaction between positively charged poly-L-lysine (PLL) and MTs was exploited.¹²² After fabricating metallic contact pads with a 10 μm gap, PLL was layered and MAP-rich MTs settled such that they straddled both contacts (Figure 7a; top). After repeated I–V (current–voltage) measurements (Figure 7a; below), the authors extrapolated the dc conductivity of a single MT to be of the order of 90 S/m. While the approach using PLL to coat a surface with MTs reduced the contribution from background buffer solution, the calculation of MT dc conductivity was based on the assumption

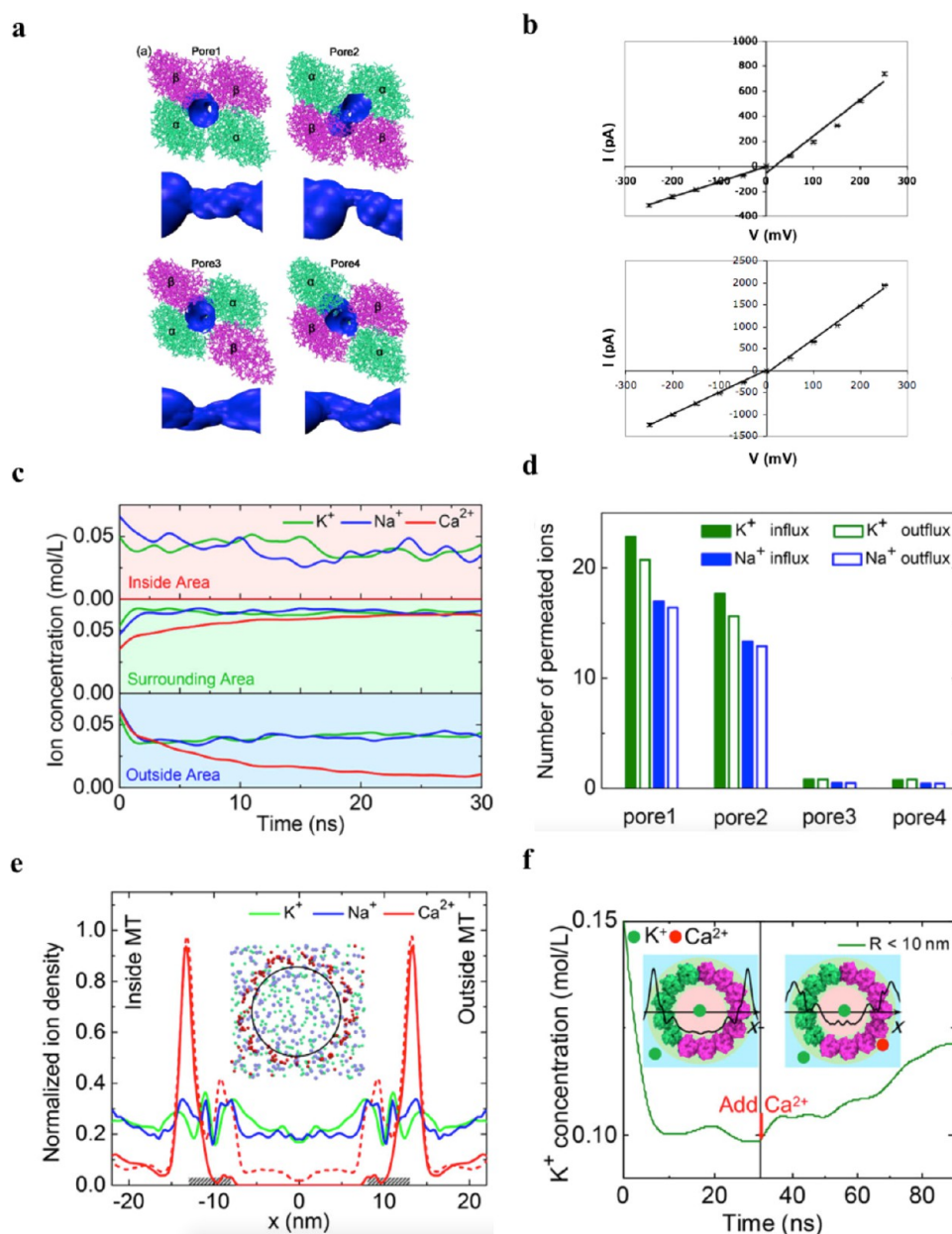


Figure 9. (a) Snapshots of nanopores created between tubulin dimers within the MT lattice. Reproduced with permission from ref 128. Copyright 2018 American Chemical Society. (b) Representation of simulated current as a function of voltage across type 1 (top) and type 2 (bottom) pores in the wall. Reproduced with permission from ref 73. Copyright 2010 American Physical Society. (c) Variation of ionic concentration of ions as a function of time, inside the MT lumen (red), outside the MT outer surface (green), and in bulk solution (blue). Reproduced with permission from ref 128. Copyright 2018 American Chemical Society. (d) Graph displaying the higher number of ions permeating across pores 1 and 2, as opposed to pores 3 and 4. Reproduced with permission from ref 128. Copyright 2018 American Chemical Society. (e) Distribution of K^+ ions around the MT wall. Reproduced with permission from ref 128. Copyright 2018 American Chemical Society. (f) Display of the influx of K^+ ions as a function of time. Reproduced with permission from ref 128. Copyright 2018 American Chemical Society.

that the PLL-MTs had adsorbed all ionic species present within the solution, and that MT resistance was connected to that of the solution in parallel. Additionally, it was unclear if the MTs denatured after several minutes of repeated voltage exposure, or if the presence of MAPs in solution altered conductivity.

To study the dc conductivity of a single MT, a dual patch clamp technique was used to directly contact the ends of an MT (Figure 7b; top). This approach avoided extrapolating ensemble MT measurements as being connected in series or parallel, thus providing for a more accurate presentation of MT dc conductivity. Interestingly, these experiments revealed an

increase in the collected current signal as opposed to background solution (BRB80; Figure 7b; bottom).¹²³ Based on differences between stimulating and collected currents at the patch pipets, it was concluded that MTs acted as ionic current amplifiers and suggested their potential roles as intracellular transistors that could regulate synaptic plasticity through electrical information transmission (ionic currents).

Measurements of MT angular velocity in an ac electric field were used to determine that MT counterionic conductivity was 150 mS/m in the 1 kHz to 10 MHz range (Figure 8a; bottom). These experiments used MTs stabilized in taxol, within a highly

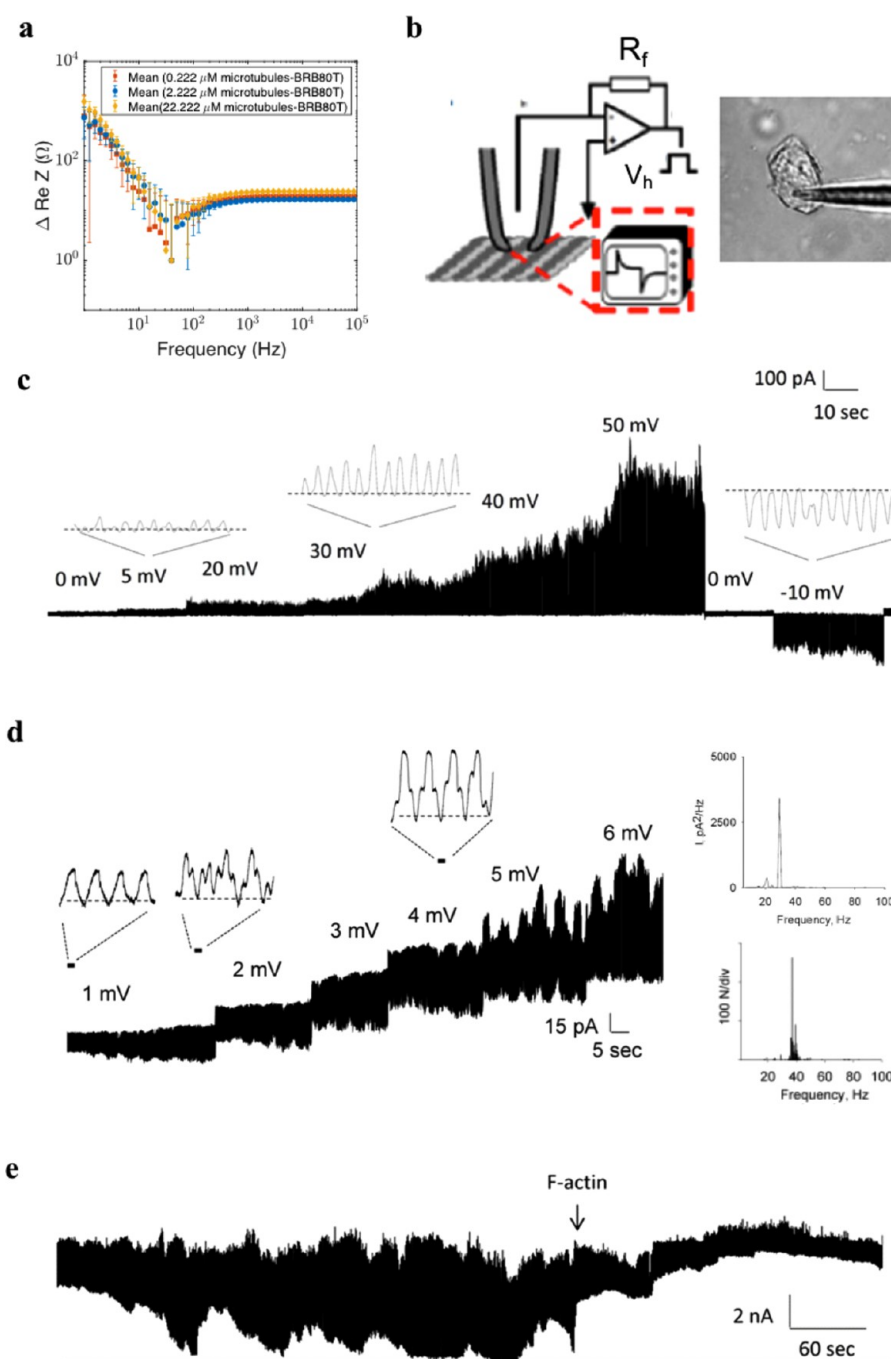


Figure 10. (a) Impedance spectroscopy measurements displaying an increase in solution conductance in the 20–60 Hz range upon the addition of MTs. Adapted with permission from ref 127. Copyright 2020 Multidisciplinary Digital Publishing Institute. (b) Patch-clamp based experimental setup to determine the current response of MTs and sheets. Reproduced with permission from ref 3. Copyright 2019 Springer Nature. (c) Oscillatory current response of an MT sheet voltage-clamped at different potentials. Reproduced with permission from ref 7. Copyright 2016 Springer Nature. (d) Oscillatory response of a voltage-clamped MT bundle as a function of time. Reproduced with permission from ref 6. Copyright 2018 Springer Nature. Insets display Fourier transforms of a typical voltage-clamped MT sheet response (upper inset) and of a typical voltage-clamped MT bundles response (lower inset). Fundamental oscillation frequencies of ~30 Hz for MT sheets and ~40 Hz for MT bundles are clearly shown. Reproduced with permission from ref 7. Copyright 2016 Springer Nature. (e) Change in the MT sheet voltage-clamped response upon addition of f-actin to the solution. Reproduced with permission from ref 4. Copyright 2020 Wiley-VCH.

diluted (“low salt”) buffer (1 mM MES; Figure 8a; top).²² When C-termini were enzymatically cleaved by the addition of subtilisin, the measured conductivity was reduced to 60% of the reference value indicating the importance of C-termini in MT-based charge transport. Adding different salts to the background solution showed that ionic conductance was highest in the presence of monovalent K^+ and Na^+ and reduced

successively in the presence of divalent and trivalent ions. In relatively higher ionic strengths (BRB16), dielectrophoresis experiments showed that at 2.5 MHz, MT conductivity was 250 mS/m.²¹

Electrochemical impedance spectroscopy using nonuniform ac electric fields (Figure 8b; top) and low ionic strength buffers (ionic concentrations ~16 mM) indicated that MTs increase

solution conductance while unpolymerized tubulin dimers decrease solution conductance at identical concentrations (Figure 8b; bottom).⁷⁶ Interestingly, the frequency range (100–300 kHz) in which this behavior was observed matched that of TTFields (tumor-treating electric fields). While such an observational correlation may indicate potential involvement of MT-based ionic transport in TTField-based tumor growth inhibition, it should be noted that these experiments were performed *in vitro* (using low ionic strength buffer solutions as background). As a consequence, these results may have limited applicability to *in vivo* systems due to the screening effects of phospholipid bilayers forming membranes. However, the screening effects are imperfect for several reasons: (a) the “leaky” capacitor membrane structures do allow for some electric field penetration through ion channels, membrane proteins and pores; (b) in dividing cells, especially during cytokinesis, the high curvature of the membrane allows a greater degree of electric field penetration than a rounded shape in interphase; and (c) there are frequency-dependent effects of electric field attenuation across the membrane with intermediate frequency values in the TTField range exhibiting lower attenuation than either higher or lower frequencies.¹²⁴

To reduce experimental noise due to the high conductance of the background buffers, which masks MT conductance, the ac experiments discussed above were performed with low ionic strength solutions (between 2 and 32 mM ionic concentration). Additionally, low tubulin concentrations were used, aiming to reduce MT number in solution and thus reduce noise and simply the modeling of an equivalent electrical circuit. However, intracellular ionic strengths are appreciably higher (200–500 mM),^{125,126} and MT networks can exist in densely packed, highly interconnected configurations within the cell. Thus, to partially mimic intracellular conditions, utilizing BRB80 (ionic strength 160 mM) and physiologically relevant tubulin concentrations, impedance spectroscopy experiments using a parallel-plate contact configuration (Figure 8c; top) were performed. It was observed that the presence of MTs caused a dramatic increase in solution capacitance while the presence of tubulin did not alter it significantly (Figure 8c; bottom).¹²⁷ Notably, when the resistance of MT-containing solution was compared to that of high ionic strength background, a “flip” in conductance was seen in the 20–60 Hz range, indicating that MTs increased solution conductance within this frequency range compared to free tubulin.

While MTs clearly introduce significant changes in conductivity and capacitance of the solution in which they are contained, it is less clear how to model the entire interconnected MT meshwork. Equivalent circuit models accounting for the electrical behavior of MTs must incorporate subtle, nonlinear aspects⁸⁸ in addition to random alignments and connections formed by MTs. Such computational modeling is important for designing and characterizing MT networks with well-controlled *I*–*V* characteristics in the ac domain.

Ionic Currents across the MT Wall. In addition to being transported along the MT outer surface, ions have been modeled to permeate across the MT wall, through gaps, or “nanopores” created between adjacent tubulin dimers (Figure 9a). There are four pore types. Type 1 and type 2 pores, which are the most common, are the intra- and interdimer gaps, with radii of 4 and 3 Å respectively, in their most constricted area. Type 3 and type 4 pores are found exclusively at the MT “seam,” forming as a consequence of the gap between two protofilaments. The

present literature indicates that MTs act as “leaky” pipes, with ions permeating into the lumen and being released at MT ends.

Cations have been modeled to permeate across the pores using energy supplied by thermally driven C-termini conformational changes. Brownian dynamics simulations of ionic conductance found that pores acted as ionic valves, with the conductance being higher when ions were moving from outside to inside the MT (Figure 9b).⁷³ When tubulin charge was removed, cationic conductance was calculated to be lowered by a factor of 15. Ionic permeation has also been modeled to arise due to the difference in electrostatic charge density between the internal and external MT surfaces.¹²⁸ These MD (molecular dynamics)-based simulations also found that type 1 pores were more conductive than type 2 pores (Figure 9c, d). Although K⁺ and Na⁺ ions permeated across the MT wall, Ca²⁺ and Mg²⁺ ions were unable to do so, due to their higher electrostatic attraction to the electrostatically negative external surface and their larger hydration shells (Figure 9e). Interestingly, it was seen that the concentration of K⁺ inside the lumen increased in response to a pulse of Ca²⁺ ions traveling along the outer MT surface (Figure 9f). Thus, the transfer of cations across the MT wall clearly plays a key role in the polymer’s overall electrical properties. Importantly, this work did not include C-termini tails (not resolved crystallographically due to their flexibility), because they do not contribute to the overall MT structure or nanopore dimensions. However, due to the C-termini containing a large fraction of the overall MT charge, and previous simulations showing its importance in charge transport across the MT wall, the exclusion of the C-termini is a serious drawback and could be incorporated in a future work.

Experimental validation of ionic transport across the MT wall finally came in 2016, when patch-clamping experiments on tubulin sheets showed cationic permeation across these sheets representing an MT wall (Figure 10b).⁷ The ionic current was found to be oscillatory with a fundamental frequency of ~29 Hz that did not change, irrespective of the holding potential used (Figure 10c and d). The average change in conductance over an oscillation at physiological conditions (140 mM K⁺) was measured to be 640%. The presence of taxol (which binds to tubulin at an intermediate binding site within the nanopores and in the main binding site to the lumen surface of an MT) decreased the time-averaged conductivity and strongly inhibited ionic oscillations. A similar oscillatory response was observed in MT bundles, which demonstrated a slightly different fundamental frequency (typically 39 Hz, which could result from differences in nanopore dimensions), a more complicated amplitude response, and a smaller mean oscillatory swing corresponding to a 258% conductance change (Figure 10d).⁸ For both MT sheets and bundles, the mean current-to-voltage response was linear over short time periods but displayed rectification over longer times. These measurements led to the hypothesis that a nanopore behaves as an electrical oscillator and the total electrical response of an MT is a result of synchronization between the oscillations. Recent measurements have shown that single MTs also display an oscillatory response to constant holding potentials, although the frequency response is much more complex (five different fundamental frequencies, all between 1 and 100 Hz, were observed) and the changes in conductance are greater than those found for previously measured tubulin structure.⁵ Impedance spectroscopy experiments performed on MT-containing solutions validated these findings, displaying an increase in solution conductance in the 20–60 Hz range, potentially due to ionic oscillations across the

MT wall at these frequencies (Figure 10a).¹ Although experimental evidence for nanopore electrical oscillations is strong, a theoretical explanation as to on the mechanism behind this has not been provided and should be a focus of future work.

It is worth noting that the two frequency regimes of increased MT conductance correspond to two separate ionic modes: The first frequency regime is between 100 and 300 kHz, where the increased conductance is hypothesized to occur as a consequence of ionic movement along the MT length, along its outer surface. The second frequency regime is between 20 and 60 Hz, where the increased conductance is thought to be a consequence of ionic movement across the nanopores in the MT wall, possibly induced by C-termini oscillations. An experimental exploration of the third hypothesized ionic mode, that along the MT lumen, has not yet been performed and would offer an interesting insight due to the shielding effects of the MT walls. The relative proportion of ions transported along each mode (along the MT outer surface through C-termini, along the MT lumen, and across nanopores) is also unclear and should also be a focus of future work (Figure 11). The utilization of

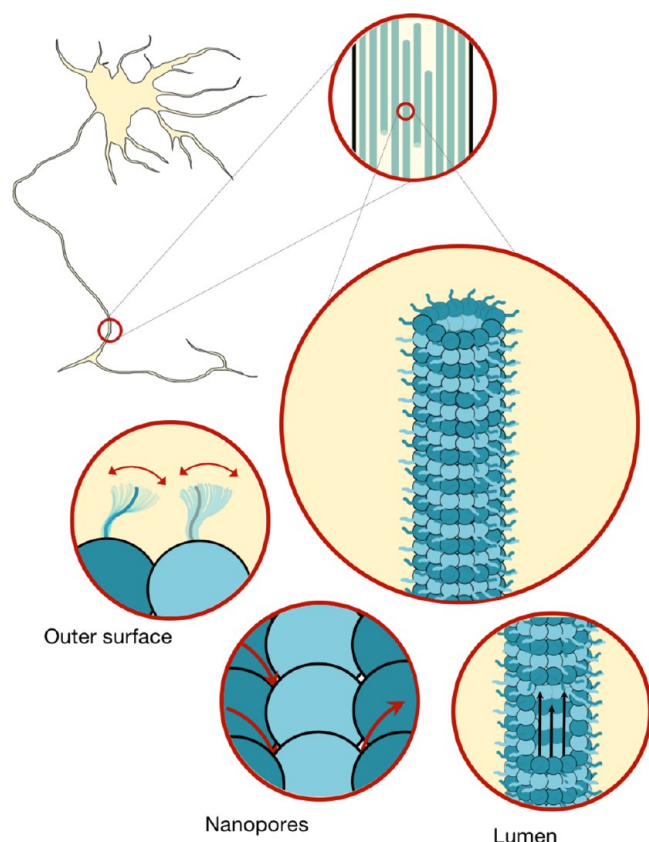


Figure 11. Demonstration of the various modes of ionic transport through MTs. Ionic transport can take place on the MT outer surface, across nanopores, and inside the lumen. Ionic signaling across MTs is modeled to play an important role in intraneuronal information processing.

TBAs to block/enhance a particular mode should be useful in this endeavor. For example, utilization of the protease subtilisin, which cleaves C-termini,¹²⁹ will nullify the C-termini's significant contribution to ionic transport along the MT outer surface. To determine the proportion of ion flow through the MT lumen, 2D tubulin sheets, which do not have a lumen, and hollow, >200 nm wide "macrotubes" can be separately subjected

to patch clamping and impedance spectroscopy, allowing for greater transport through the inner MT circumference. It is unclear how ionic transport will scale with dimensions of these polymorphic structures, given the presence of "leaks" through nanopores and the different geometries. Nevertheless, experiments performed on sheets and macrotubes should shed light on the contribution of the MT lumen to ionic transport. Future experimental research must utilize MT-TBA interactions to decipher the relative importance of each mode. Notably, much of our present knowledge of MT electrical behavior is based on experiments performed in the presence of taxol (which inhibits ionic transport through nanopores^{3,6,7}). By comparing the impedance parameters of taxol-stabilized MTs to those that are stabilized by a TBA that does not block nanopores (such as the MAP TPX2¹³⁰), it should be possible to glean information on the contribution of nanopores to MT-based ionic transport. Understanding how TBAs, actin filaments and cell organelles alter the ionic impedance around MTs will also enable us to understand their role in intracellular MT-based electrical signaling. This would enable a construction of an effective bioelectric circuit of a living cell.

Early work in this direction already indicates that the electrical behavior of MT nanopores changes in the presence of actin filaments,⁴ previously only known to interact with MTs biochemically through MAPs such as MAP2 and MAP tau.^{131,132} In patch-clamp measurements, the presence of actin filaments led to a significant increase in the mean current across MT sheets while keeping their oscillatory frequency unchanged (Figure 10e). A similar response was observed for MT bundles where the charge transfer increased by about 100% after the addition of actin filaments. Electrostatic "crosstalk" between actin filaments and MTs may have biological implications, such as the generation of propagating ionic soliton waves around and along actin filaments in the neighborhood of MT walls that generate such ionic oscillations, which could be of significance in excitable cells such as neurons and muscle cells where actin filaments are involved in neurotransmitter release and in force generation, respectively.^{89,133} Understanding how each impedance component varies with MT length is critical for the fabrication of future MT-based electrical devices. Due to the presence of three separate modes of ionic transport in a single MT, addressing such issues is nontrivial. The significance of "leakage current" along the MT length is also unclear and should be included in future models. Understanding the variation of ionic conductivity with MT diameter should give insights into the importance of leakage current, being critical from both a device perspective as well as an evolutionary standpoint. More sophisticated computational models and future experiments should, therefore, consider all three modes of ionic transport and changing MT dimensions and include interactions with actin filaments, tubulin-binding drugs, and MAPs.

Due to their exotic electrical properties, MTs have been modeled as "memristors" (two-terminal devices that are described by a resistance which dependent on the previous conductive state of the device, *i.e.*, resistance with memory).¹³⁴ Changes in the C-termini conformation have been theorized to affect changes in MT resistance to counterionic currents, since the C-termini conformation is influenced by the counterionic concentration.^{1,2} This results in the emergence of memristance, which has been calculated to be between 1 Ω and 20 k Ω for a 10 μm MT. Although direct experimental proof of these results is still required, it was found that, when subjected to the same magnitude of dc voltage, a solution containing MTs responded

with different resistance values depending on the direction of the voltage, which is consistent with this assumption.¹³⁵ It was also seen that when patch-clamp based square-voltage pulses were applied to “silent” tubulin sheets (those that did not display oscillatory responses under voltage-clamp conditions), the response could be modeled by a circuit with a saturating capacitor and a memristor. While more work (experimental and theoretical) is required to verify such extraordinary and far-reaching claims, memristive properties of MTs suggest many intriguing possibilities. It has been shown that memristors can implement logical operations¹³⁶ and they have great promise for use in neuromorphic computing.¹³⁷ Tuszynski *et al.* tie these findings back to basic biology and suggest that MT memristive properties may be behind the “memory” demonstrated by amoeba.¹ Thus, memristive properties of MTs would have extensive implications for biocomputing applications and basic biological research. Moreover, due to the prominent role played by MTs in the neuronal cytoskeleton, where they form parallel bundles interconnected by MAPs, their potential involvement in memristive signal processing may be of major importance to our understanding of how the human brain works. The sheer complexity of the electrical properties of MTs and their sensitivity to the ambient conditions may offer a rich spectrum of possibilities for signal processing in both living systems and their hybrid technological adaptations.

PERSPECTIVES

In addition to communicating through biochemical pathways, intracellular macromolecules communicate through bioelectrical cues. Both theoretical/computational modeling and experimental work illustrate the significance of MTs in such communication. MTs can respond to bioelectrical signals through mechanical migration,^{9,10,75,82} electrorotation,^{22,87} changes in C-termini conformation,^{119,120} mechanical stiffness,⁹⁵ and surface charge,¹⁷ and ionic transport along the outer surface¹²³ and across the nanopores.^{3,6,7} These electrical, biochemical, and mechanical responses are, of course, interrelated, with the precise boundaries of each response regime being largely unclear. While significant advances in our understanding of MT response to electrical stimuli have been achieved in the past half-decade, further work must be performed to determine the exact nature (electric field intensity, frequency, and waveform) of the electrical stimuli that elicit each response. Future work should focus on not only how electric fields alter the interactions of MTs with TBAs such as kinesins but also how MT–TBA interactions alter MT electrical properties. Careful research on the interplay between external electrical field application, TBA binding and resultant MT properties is needed to fully realize the potential of MTs as tunable electrical conduits and ionic transporters.

Characterization techniques will need to be advanced to obtain quantitative information about MT electrical properties. The concentration of various cations in the immediate vicinity of MTs is useful to know in this connection. Currently, there are no reports on the empirical measurement of local ionic concentrations for tubulin dimers and MTs. As previously mentioned, our understanding of the different modes of conductance in a MT is largely incomplete and, in particular, the nanopores in MTs have received little attention (Figure 11). The binding of fluoroionophores to select amino acids in MTs might offer a way forward to estimate the local concentrations of specific cations through fluorescence and the changes in the local concentration near nanopores to obtain insight into the

ionic conductance of the nanopores. For instance, potassium fluoroionophores based on dye-conjugated crown ethers and cryptands are able to selectively detect K⁺ ions (even in a Na⁺ rich electrolyte)¹³⁸ and 4-pyridylethenylbenzene derivatives can detect local proton concentrations through shifts in their emission spectra.¹³⁹ Patch clamp techniques are routinely used to study the electrical properties of protein nanopores.¹⁴⁰ Advanced scanning probe microscopy (SPM) techniques tailored to nanopores such as scanning ion conductance microscopy constitute a powerful suite of high spatial resolution characterization techniques to probe the conductance signatures of type 1 and type 2 nanopores in MTs.¹⁴¹ One challenge associated with nanopore SPM is the interpretation of the conductance signatures obtained.¹⁴² The ability to make direct electrical contact to the end of MTs through lithographically defined electrodes would enable gating of the nanopores in MTs to modulate the ionic conductance akin to a transistor.¹⁴³ Even the ion-sensitive field-effect transistor (ISFET) configuration commonly used in biosensing offers opportunities to advance the characterization of MTs. If the gate dielectric of an ISFET was constituted of a single layer carpet of MTs or zinc sheets, the saturation voltage and charging gate current during gate voltage sweeps could yield information regarding the total charge and conductance of nanopores in MT ensembles respectively. Although electric fields have been reported to align MTs, we are not aware of a mat of fully aligned MTs. Such a carpet of aligned MTs straddling two metal electrodes would enable magnetoresistive probing of the nanopore conductance in the alignment direction (orthogonal to the axes of the MTs) through the Hall effect by placing a third electrode at the side of the aligned carpet.

From a fundamental science standpoint, a better understanding of the role played by the lumen and C-termini for MT-based charge transport is crucial to both the fabrication of MT-based nanodevices and the determination of the electric properties of living cells. While deep inroads into our understanding of MT bioelectricity have been achieved by investigating their mechanical response to electrical stimuli, future utilization within nanodevices and medical modalities will require an improved understanding of their electrical response.

We now know that the electrical behavior (conductivity, surface charge) of free unpolymerized tubulin in solution is dramatically different from that in its polymerized forms.^{1,76} However, understanding how the electrical properties of free tubulin change as it polymerizes will help reveal how MT length and sheet dimensions control their electrical behavior. Future work showing precisely how actin filaments and TBAs alter MT electrical properties should shed further light on the promise of the cytoskeleton as a tunable, dynamic, and sensitive electromechanical signal transduction device.¹⁴⁴ Indeed, the effect of MT electromechanical properties at the cellular system level must be explored. Two-dimensional models have already displayed an enhancement of the piezo- and flexoelectric properties of the cell upon cytoskeletal degradation.¹⁴⁵ Expanding on such work will require a greater understanding of how to couple MT electromechanical properties to energy sources including acoustic waves. Recent work has also shown the importance of tunneling nanotubes (TNTs), which are membranous bridges formed between neighboring cells, for the transport of intracellular organelles and cytoplasmic molecules.^{146,147} Interestingly, TNTs, which can contain MTs,^{148,149} also transmit electrical signals through calcium waves between cells.^{150,151} Given that MTs are modeled to play important

Table 2. List of 10 Challenges for MT Bioelectricity

field of impact	duration	question
intracellular biophysics	short-term	How do electric fields interact with dynamic instability? How to electric fields alter MT nucleation, elongation rate, shortening rate, catastrophe frequency and rescue frequency?
intracellular biophysics	short-term	Do MTs preferentially transport ions? Which ions are transported preferentially?
intracellular biophysics	medium-term	How do MAP and drug binding, and the presence of actin filaments, alter MT conductivity?
intracellular biophysics	long-term	Does MT/tubulin polymerization state alter the local electrical properties of the cell? Does MT shortening alter the local spatiotemporal electrical properties? Can shortening of one MT trigger intracellular events driven by this electrical change?
intracellular biophysics	long-term	What is the ratio of ionic transport among these three modes? 1. Lumen transport 2. Outer surface transport 3. Nanopore transport How to distinguish between these conductance modes experimentally?
intracellular biophysics + nanodevice applications	short-term	How do MT-MAP interactions change when exposed to electric fields? Do electric fields cause biochemical changes in the cytoskeleton?
nanodevice applications	medium-term	How do MTs interact with acoustic waves? Can MT electrical properties be used for electromechanical coupling?
nanodevice applications	medium-term	How does conductance depend on MT length and diameter? Unclear due to the presence of nanopores and ionic "leakage" through outer surface.
intracellular biophysics + medical applications	medium-term	What is the effect of insulating membranes on MT coupling to ac electric fields? How does the presence of pores in these membranes influence MT coupling? How do these results correlate to the effects of ac electric fields acting on living cells?

roles in the storage and transport of intracellular ions, the relationship between ionic transport by MTs and that by TNTs must be explored. It would also be critical, from a medical perspective, to investigate if the characteristics of ionic transport are altered in diseased tissue, particularly brain tumors which have been shown to contain thicker, longer intercellular bridges termed tumor microtubules (TMs).¹⁵² Further, the effect of electric fields, particularly TTFs, on TNT-related MT transport would be interesting to study.

Outside the cell, if devices using tubulin and its polymers are to be fabricated, the electrical and mechanical properties of these biological polymers in solvents such as DMSO and DMF, which are commonly used for the fabrication of OLEDs and OPVs (organic photovoltaics), must be precisely determined.¹⁵³ Experiments performed in a variety of solution conditions (over a variety of solvents, pH values, and ionic strengths) may reveal the origin of differences in reported values of conductivity and effective charge. As discussed in previous sections, theoretical models have postulated that MTs behave as transmission lines capable of propagating nonlinear ionic waves^{119–121} and that MTs have memristive properties.¹ Initial experiments have supported these hypotheses and shown that MTs can amplify propagating ionic signals.^{3,74} These results suggest that MTs could be a building block of bioelectronic circuits and biocomputational systems. To verify and develop this possibility, further experimental quantification of MT electrical properties is required. In addition to creating hybrid bio-nanoelectronic devices, implantable subcellular structures based on MTs can be engineered for applications such as neuroregeneration, memory augmentation, or targeted drug delivery.^{154–156} If the challenges outlined above can be overcome, then the potential of MTs to reduce the complexity of microelectromechanical systems, while simultaneously enhancing their versatility and functionality, is monumental and can create avenues for a wide spectrum of applications.

We close this work with a list of 10 questions (Table 2) that must be answered in the field of MT bioelectricity in order to rapidly advance in the quest outlined above. Successfully addressing these questions will improve our understanding of MTs within the cell while also enabling their utilization within medical and possibly bioelectronic devices.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

All authors thank NSERC and Novocure (Haifa, Israel) for financial support. A.P.K thanks Alberta Innovates for graduate student scholarship funding. A.P.K thanks Mahshad Moshari, Prof. Travis Craddock, Dr. Holly Freedman, Dr. Jordane Preto, and Philip Winter for useful discussions. B.B.E thanks the Alberta Innovates Summer Research Studentship for financial support. A.P.K dedicates this review to Prof. P. S. Satsangi.

VOCABULARY

Persistence length, the distance over which the two-point correlation function for the vector tangent to the arc length decreases exponentially; **persistence length** of a polymer is proportional to its bending stiffness; **Zeta potential**, the electrostatic potential at the slipping plane of a nanoparticle; the space between the nanoparticle and the slipping plane consists of counterions that are immobilized (“frozen”) at the nanoparticle surface; **Bjerrum length**, the distance at which the electrostatic energy between two electrostatic particles is equivalent to the thermal energy; **Dielectrophoresis**, the movement of an electrostatically neutral but polarizable particle induced by the application of a nonuniform electric field; **Memristor**, a two-terminal device characterized by a resistance that depends on the previous conductive state of the device, *i.e.*, resistance with memory

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